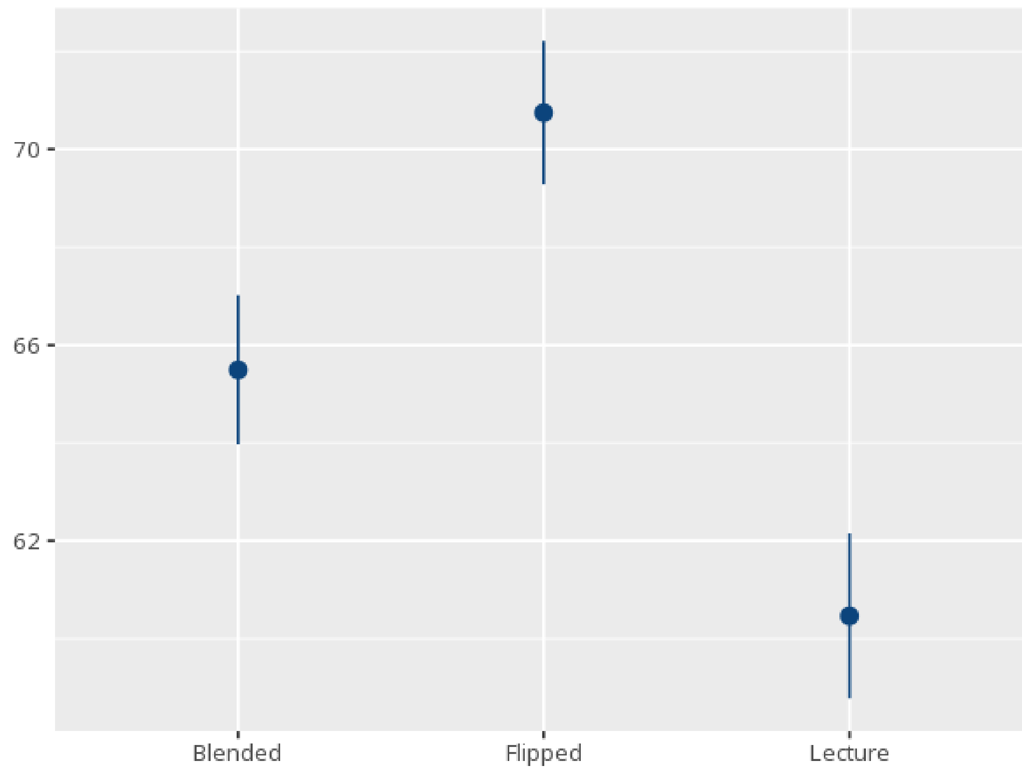
1 Welch's ANOVA

Group	N	Mean	Std. Dev.
Blended	80	65.49	6.85
Flipped	80	70.75	6.58
Lecture	80	60.46	7.57

F	df1	df2	p	ω^2 [95% CI]
42.31	2	157.49	$<.001$	.34 [.24, 1.00]

Pair	Mdiff	padj	95% CI
Blended - Flipped	-5.26	$<.001$	$[-7.77, -2.75]$
Blended - Lecture	5.03	$<.001$	[2.33, 7.73]
Flipped - Lecture	10.29	$<.001$	[7.63, 12.94]



A one-way ANOVA that relaxes the equal-variance assumption but still assumes roughly normal data within each group. A significant F means at least one group mean differs from the others; the Games-Howell post-hoc (also variance-robust) then shows which specific pairs differ. Your significance threshold (α) is 0.05.

Welch's ANOVA gave F(2,157.49)=42.31, $p < .001$, ω^2=.34 [.24, 1.00].

Statistical basis: Welch's ANOVA (heteroscedasticity-robust F, fractional df) (Welch, B. L. (1951). "On the comparison of several mean values: an alternative approach." Biometrika, 38(3-4), 330-336.); Games-Howell post-hoc comparisons (also variance-robust) (Kassambara A (2023). rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package.); ω^2 effect size (F-to-omega^2 conversion, no fitted aov model needed) (Ben-Shachar MS, Lüdecke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." Journal of Open Source Software, 5(56), 2815.).